

Portfolio Assignment Briefing-Report

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Section : 10

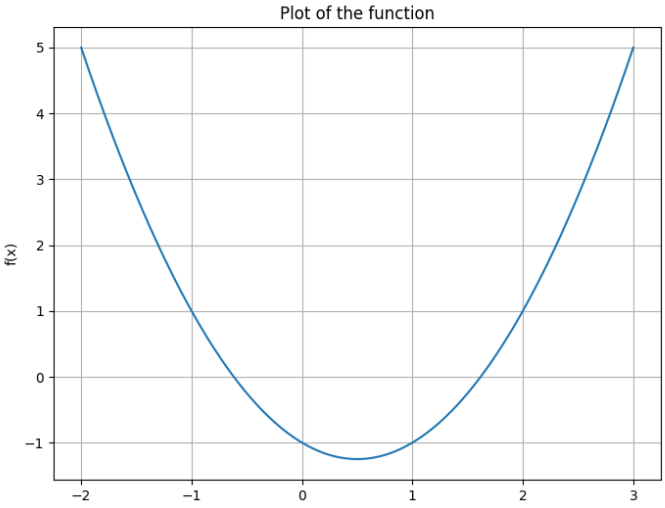
Submitted to : Nabin Acharya (Tutor)
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Task-1

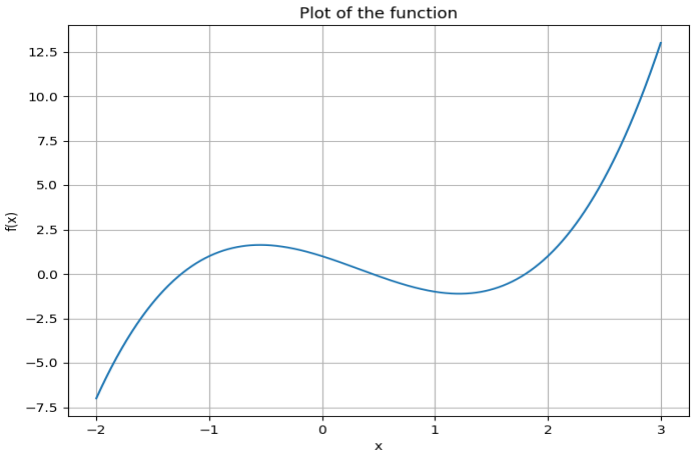
Bisection method

Plot of the functions:

<u>For function 1</u> (y = f(x) = x^2 – x - 1)



<u>For function 2</u> (y = f(x) = x^3 – x^2 – 2x + 1)



The Bisection technique formula is:

(A + B) / 2 equals c.

Using this method, a and b stand for the interval's beginning and ending positions, respectively, and c for the interval's midpoint. When $f(c)$ becomes closer to zero, it means that c is getting closer to the root. Each time, c is modified, substituting either a or b, based on how $a(c)$ is compared to $a(a)$ and $f(b)$.

Rate of convergence of the Bisection Method: By progressively shrinking the interval surrounding the root, the Bisection Method can converge. This procedure guarantees a reduction in error, which is the discrepancy between a root's true value and an approximation. The accuracy bound formula's concept of i^2 illustrates how errors go smaller with each repetition. The value of 1^2i gets less and smaller as the number of iterations (i) increases.

Formula of error: $|x_i - c| \leq (b_i - a_i) / 2$
We know that $(b_i - a_i) / 2 = (b - a) / 2^i$
So, $|x_i - c| \leq (b - a) * 1 / 2^i$

Therefore, it's order of convergence is 1

Formula for Newton Raphson method:

$\boldsymbol{X_{n+1} = X_n - f(X_n)/f'(X_n)}$ where $\boldsymbol{f(X_{n+1})}$ must be 0.
where n is the current point and n+1n+1 is the next point that needs to be found.

Convergence rate of the Newton Raphson technique
With each iteration, the Bisection method's rate of convergence grows quadratically, exhibiting quadratic convergence. When the square of $f'(x)$ is less than the absolute value of the product of $f(x)$ and $f''(x)$, convergence is reached. On the other hand, if $f'(x) = 0$, then the procedure fails.

Consequently, its order of convergence is determined to be 2.

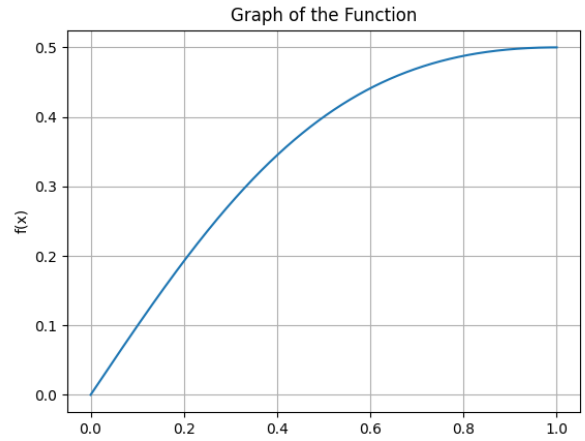
For Function 1			
Bisection Method		Newton's Method	
$[a, b]$; #iterations; Root1	$[a, b]$; #iterations; Root2	$x0$; #iterations; Root1	$x0$; #iterations; Root2
[1,2]; #16; 1.6180343627929688	$[-2,0]$; #17; -0.6180343627929688	1.75; #3; 1.61803399	-0.9; #3; -0.61803404
SciPy Method			
$x0$; Root1		$x0$; Root2	
1.75; 1.61803399		-0.9; -0.61803399	

For Function 2					
Bisection Method			Newton's Method		
$[a, b]$; #iterations; Root1	$[a, b]$; #iterations; Root2	$[a, b]$; #iterations; Root3	$x0$; #iterations; Root1	$x0$; #iterations; Root2	$x0$; #iterations; Root3
$[-2, -1]$; #21; -1.2469794750213623	[0.3,0.6]; #18; 0.44504184722900386	[1.80,1.81]; #13; 1.8019378662109378	-1.5; #4; -1.24697960	-0.25; #4; 0.44504182	1.5; #5; 1.80193774
SciPy Method					
$x0$; Root1		$x0$; Root2		$x0$; Root2	
-1.5; -1.24697960		-0.25; 0.44504187		1.5; 1.80193774	

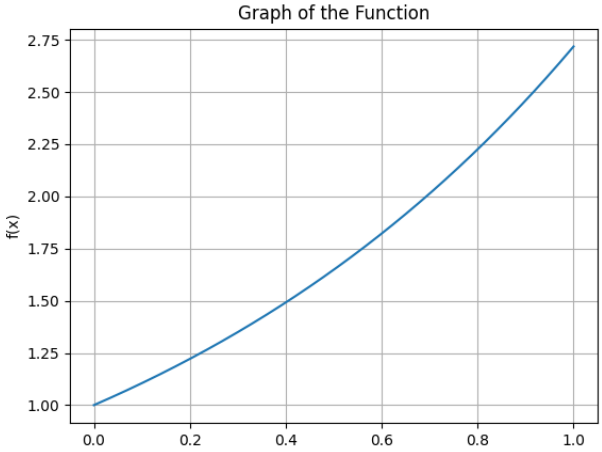
Task-2

1.1 Plot of the functions

Function 1 ($y = f(x) = \frac{x}{x^2+1}$ **)**



Function 2 ($y = f(x) = e^x$ **)**



Formula for Midpoint Approximation:

$$h = (b - a)/N$$

$$Area \approx \sum_{i=1}^{n} f(m_i) * h$$

where, h represents the width of the sub-intervals, and mi signifies the value of x at the midpoints

Formula for Trapezoidal Rule:

$$h = (b - a)/N$$

$$Area = \frac{h}{2} * (f(x_0) + 2f(x_1)) + 2f(x_2) + \cdots + f(x_n))$$

Where h is the width of the subintervals, and f(x) means the functional value of x at each of the partitions.

N	Midpoint Approximation (M_N)	Trapezoidal Rule (T_N)	Analytical (I)	Abs Error $\lvert M_N - I \rvert$	Abs Error $\lvert T_N - I \rvert$
For Function 1					
10	0.346991	0.345739	0.346574	0.00041758	0.00083438
30	0.346620	0.346481		0.00004631	0.00009261
50	0.346590	0.346540		0.00001667	0.00003334
100	0.346578	0.346565		0.00000417	0.00000833
500	0.346574	0.346573		0.00000017	0.00000033
For Function 2					
10	1.717566	1.719713	1.718282	0.00071574	0.00143166
30	1.718202	1.718441		0.00007955	0.00015910
50	1.718253	1.718339		0.00002864	0.00005728
100	1.718275	1.718296		0.00000716	0.00001432
500	1.718282	1.718282		0.00000029	0.00000057

Error percentage of Trapezoidal rule:

= (Actual−Approximated) / Actual ×100 %

For function 1, error when N=100

Error = (0.346574−0.346573)/ 0.346574*100%

= 0%

For function 2, error when N=100

Error = (1.718282−1.718282)/ 1.718282*100%

=0%

Because a high number of partitions (100) were used, the error in both solutions derived by the application of the Trapezoidal rule is 0% when compared to the analytical solution. This made sure that the divisions shrank enough so that the areas of the trapezoids could be added together precisely and without error.

Error percentage of Midpoint rule:

$=(\text{Actual}-\text{Approximated}) / \text{Actual} \times 100\ \%$

For function 1 , error when N=500

Error = (0.346574-0.346573)/ 0.346574*100%

= 0.00000288538%

For function 2 , error when N=500

Error = (1.718282-1.718282)/ 1.718282*100%

=0%

Alternatives to finding function roots besides the bisection and Newton's methods include the secant method and fixed-point iteration. These alternatives do not require calculating derivatives, making them simpler to use compared to Newton's method which does require derivatives. The secant method and fixed-point iteration generally converge faster than the bisection method, with the secant method being the quickest. However, the secant method requires two initial guesses and may not always find a solution, while fixed-point iteration can converge slowly or fail to find a solution depending on how it is set up. The choice between these root-finding methods involves trade-offs that depend on the specific situation.

For estimating the area under a curve, alternatives besides Simpson's rule and Monte Carlo integration include the trapezoidal rule and midpoint rule. Monte Carlo can handle complex integrals but is computationally expensive. Simpson's rule is accurate for smooth functions but requires evenly spaced intervals. The trapezoidal and midpoint rules are simpler but slightly less precise.

In summary, the bisection and Newton's methods are effective root-finders, with Newton's being faster but needing derivatives. The trapezoidal and midpoint rules estimate areas, with midpoint preferable for smoother functions. Method selection depends on the problem's characteristics. Generally, methods requiring fewer iterations to approximate the solution have better, faster convergence rates.